

```

C:\py3.7>python pyinstaller.py dns1.py --onefile
793 INFO: PyInstaller: 3.4
793 INFO: Python: 3.7.3
793 INFO: Platform: Windows-10-10.0.17134-SP0
808 INFO: wrote C:\py3.7\dns1\dns1.spec
808 INFO: UPX is not available.
808 INFO: Extending PYTHONPATH with paths
['C:\py3.7', 'C:\py3.7\dns1']
808 INFO: checking Analysis
808 INFO: Building Analysis because Analysis-00.toc is non
istent
808 INFO: Initializing module dependency graph...
808 INFO: Initializing module graph hooks...
824 INFO: Analyzing base_library.zip ...
4687 INFO: running Analysis Analysis-00.toc
4702 INFO: Adding Microsoft.Windows.Common-Controls to depe
ent assemblies of final executable
required by C:\python3\python.exe
5404 INFO: Caching module hooks...
5473 INFO: Analyzing C:\py3.7\dns1.py
5551 INFO: Processing pre-find module path hook distutils
9545 INFO: Processing pre-safe import module hook six.mov
10963 INFO: Loading module hooks...
10963 INFO: Loading module hook "hook-cryptography.py"...
11354 INFO: Loading module hook "hook-distutils.py"...
11354 INFO: Loading module hook "hook-encodings.py"...
11432 INFO: Loading module hook "hook-lib2to3.py"...
11447 INFO: Loading module hook "hook-pydoc.py"...
  
```

Figure 2: Converting Python code to an .exe file

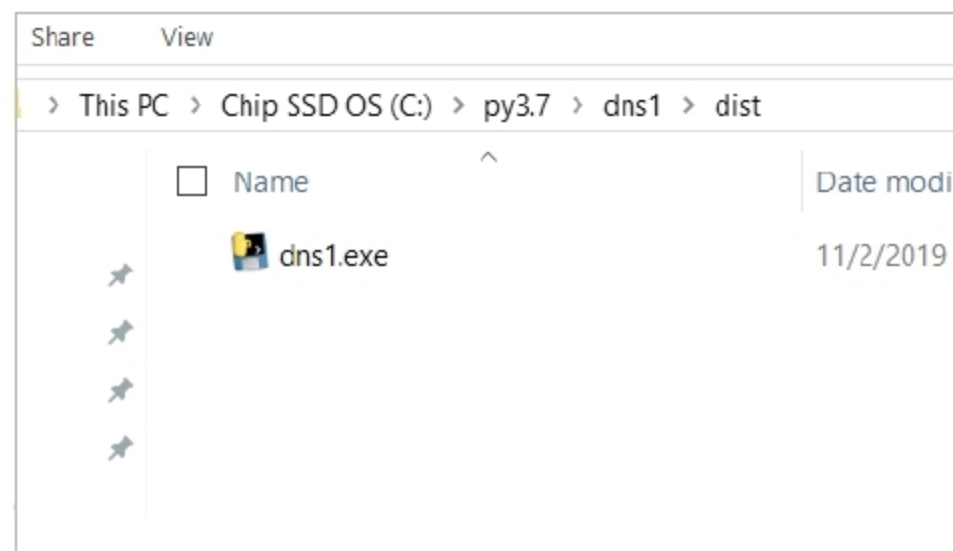


Figure 3: .exe file path

.exe, which can be run using Python. To convert Python code to .exe, we will use the Pyinstaller module, which I have at path C:\py3.7. Now copy the code to the directory C:\py3.7 and run the following command:

`Python pyinstaller.py dns1.py -onefile`

For more help, refer to Figure 2. After successfully running the command you can find the exe file at the path specified in Figure 3.

```

C:\Users\Mohit>dns1
Use -h for the options
For more help email to mohitraj.cs@gmail.com

C:\Users\Mohit>dns1 -h
usage: dns1 [-h] [-o 0 0] [-v]

optional arguments:
  -h, --help  show this help message and exit
  -o 0 0      First give pattern then IP
  -v          Version Number

C:\Users\Mohit>dns1 -o workstation 172.16.2
  
```

Figure 4: DNS server is running

```

Microsoft Windows [Version 10.0.17134.1006]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Mohit>nslookup
Default Server: UnKnown
Address: 192.168.43.1

> server 127.0.0.1
Default Server: localhost
Address: 127.0.0.1

> workstation123.mohit.com
Server: localhost
Address: 127.0.0.1

Non-authoritative answer:
DNS request timed out.
  timeout was 2 seconds.
Name:   workstation123.mohit.com
Address: 172.16.2.123

> workstation123.raj.com
Server: localhost
Address: 127.0.0.1

Non-authoritative answer:
DNS request timed out.
  timeout was 2 seconds.
Name:   workstation123.raj.com
Address: 172.16.2.123

> workstation12.raj.com
Server: localhost
Address: 127.0.0.1

Non-authoritative answer:
DNS request timed out.
  timeout was 2 seconds.
Name:   workstation12.raj.com
Address: 172.16.2.12
  
```

Figure 5: DNS queries

Now copy the exe file *dns1.exe* to the C:\Windows folder so that it can be used as a command. Next, open the command prompt and execute the command as shown in Figure 4. Your code in the form of exe is running and ready to serve as the dynamic host

resolver. Let us fire the DNS queries with the help of NSlookup. Figure 5 shows the queries and how to get the desired responses. In this figure, the red arrows indicate the query typed by the user, and the corresponding IP addresses in rectangles are returned by the Python code. **END**

By: Mohit

The author has CEH and ECSA qualifications, and has completed a Masters in Engineering (ME) in computer science from Thapar University, Patiala. He has worked in IBM, Teramatrix and Sapient. He is the author of the following books: 'Python Penetration Testing Essentials', 'Python: Penetration Testing For Developers' and 'Learn Python In 7 days'.